

Abdullah Naveed

Ph.D. Student in Computer Engineering | University of Florida
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EDUCATION

University of Florida

Aug 2025 – Present

Ph.D. in Computer Engineering | GPA: 3.93/4.00

University of Iowa

Aug 2023 – Aug 2025

Master of Computer Science | GPA: 3.99/4.00

Lahore University of Management Sciences

Aug 2019 – May 2023

B.S. in Computer Science, with Distinction | GPA: 3.61/4.00

Lahore, Pakistan

RESEARCH EXPERIENCE

Argonne National Laboratory

Jun 2025 – Present

Visiting Graduate Researcher (Remote)

Lemont, IL

- Designed an end-to-end **hardware–software co-design framework** for FPGA/CPU scientific data compression, achieving up to **19.8× cycle speedup** over matched software and **46–65% lower end-to-end workflow time**.
- Developed a **modular hardware compression library** for **constraint-driven composition** of reusable Chisel/RTL blocks into system-optimized compressors, supporting LZ4, Snappy, StreamVByte, SZ3, SZp, SZx, zfp, and extensible new designs.

University of Florida

Aug 2025 – Present

Graduate Research Assistant, Dependable Systems Lab

Gainesville, FL

- Developed **TRACE**, a runtime detector for silent data corruption in LLM inference, achieving **95.3% average detection coverage** across 15 LLMs and 3 datasets at only **0.56% runtime overhead**; paper **accepted to SC'26**.
- Conducted **large-scale LLM reliability studies** spanning **2.7M+ fault-injection trials** across 12 LLMs; evaluated **parameter-efficient LLM fine-tuning (PEFT) resilience** across **LoRA, QLoRA, LoRA-FA, and DoRA** to identify vulnerable model components and quantify performance degradation under faults.

University of Iowa

Aug 2023 – Aug 2025

Graduate Research Assistant

Iowa City, IA

- Built **LLVM-based fault-injection infrastructure** for C/C++ scientific applications and co-developed a **compiler-based SDC detector** achieving **86.19% average coverage at 4.90% runtime overhead** across 15 benchmarks.
- Characterized **data- and control-flow error propagation** and evaluated **selective runtime monitoring** against instruction-duplication baselines.

SELECTED PUBLICATIONS

- Abdullah Naveed**, Mahina Sheikh, Guanpeng Li. *Low-Overhead Silent Error Detection for Large Language Model Inference at Scale*. SC'26. **Accepted; to appear**.
- Darren Ng, **Abdullah Naveed**, Duo Zhang, Guanpeng Li, Sheng Di, Xiaoyi Lu. *Z-SPDK: Accelerating Lossy Compressed I/O on NVMe-SSDs*. SC'26. **Accepted; to appear**.
- Muhammad Hassan, Talha Tahir, Muhammad Farrukh, **Abdullah Naveed**, Anas Naeem, Fareed Zaffar, Fahad Shaon, Ashish Gehani, Sazzadur Rahaman. *Evaluating Container Debloaters*. IEEE SecDev, 2023.

MANUSCRIPTS UNDER REVIEW

- LLM Pretraining Reliability** — large-scale characterization of soft-error effects during Transformer pretraining.
- LLM Fault Resilience** — characterization of LLM behavior under increasing fault intensity and multi-bit corruption.
- HPC Reliability** — compiler-based, low-overhead silent error detection in scientific and HPC applications.

TECHNICAL SKILLS

Programming : Python, C/C++, Bash, SQL, Git

AI / Machine Learning : PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn, NumPy, Pandas

Systems / HPC : LLVM, Linux, Docker, MPI, OpenMP, Slurm, fault injection, parallel computing

Hardware : Chisel, Verilog/RTL, RISC-V, Chipyard, Verilator, Vivado